Internship report: Exploring the chromospheric emission and photometric variability of low-activity main-sequence stars

Emma Carli - University of Glasgow

Laboratoire d'Astrophysique de Marseille

Groupe Systèmes Planétaires (LAM/GSP)

10 week internship in Summer 2018

Abstract

The $\log R'_{HK}$, a measure of ultraviolet chromospheric emission, is used as a stellar sample selection tool in exoplanet surveys to reduce activity-induced noise. It has indeed been found that above $\log R'_{HK} = -4.7$, the Vaughan-Preston gap, photospheric activity increases quite linearly with chromospheric emission [Radick et al., 1998]. In this project, the validity of the $\log R'_{HK}$ as a photospheric activity indicator below this gap is explored. For this purpose, stellar spectroscopy and photometry are retrieved simultaneously on a stellar sample considered mostly inactive. The $\log R'_{HK}$ is computed in a parallel study, Vivien [2018], while this project focuses on retrieving the amplitude of variability of the stellar light curves. Autocorrelation Functions, RMS and Gaussian Processes techniques are performed on the data and their relevance is evaluated. It is found that in the low activity range, the relationship between amplitude of variability and $\log R'_{HK}$ is quite spread, although an increasing trend remains. This casts doubt on the usefulness of the indicator, and calls for a reassesment of chromospheric and photospheric activity correlation.

Contents

1 Background and technique **3**
1.1 Stellar activity and exoplanet detection 3
1.1.1 Photometry 3
1.1.2 Spectroscopy 4
1.2 Data: K2 photometry 4

2 The frame of work **5**
2.1 Large sample 5
2.2 Small sample 7

3 The project **8**
3.1 Computing rotation periods with the ACF method 9
3.1.1 Introduction 9
3.1.2 Data cleaning 9
3.1.3 Algorithm 11
3.2 Gaussian Processes 13
3.2.1 Introduction 13
3.2.2 Data cleaning 14
3.2.3 Algorithm 14
3.3 Results and analysis 15
3.3.1 Periods and amplitudes 15
3.3.2 Amplitude of variability and chromospheric activity indicator relationship 15
3.4 Discussion and openings 16
3.4.1 Improvements 16
3.4.2 Further research 17

4 Conclusion **17**

1 Background and technique

1.1 Stellar activity and exoplanet detection

A star's activity can be defined by how much its magnetic field will inhibit or enhance convection in the photosphere, leading to starspots and faculae, respectively (Figure 1). Activity causes variations in the stellar light curve and spectrum. Indeed, starspots and faculae increase and decrease stellar light flux, and both increase local redshift [Dumusque et al., 2014].

Exoplanets can be detected by stellar radial velocity measurements in spectroscopy, which determine the planet's mass, or with transits in photometric light curves that establishes radii. Both techniques can be contaminated by stellar activity features.

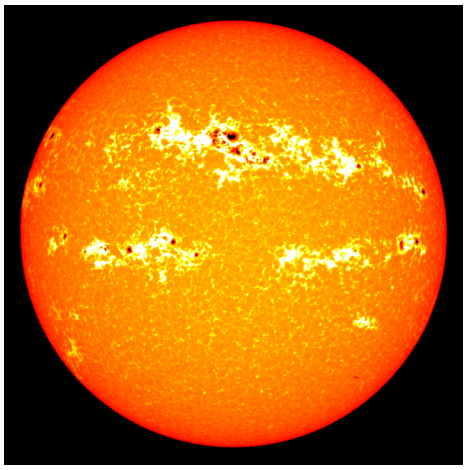

Figure 1: Groups of spots surrounded by faculae on the solar photosphere (NASA/SORCE/SVS).

1.1.1 Photometry

In photometry, light curves are affected by stellar activity. Variations in flux arise due to:

- the appearance and decay of features;
- stellar rotation, which takes the features in and out of the field of view.

By proposing an appropriate model therefore, one can infer the lifetime of the features and the rotation period of a star from its light curve (Figure 2).

For the study of transits, the activity-induced flux variations create unwanted trends and jitter in the light curves that need corrected (Figure 3). Spots cause flux variations in the order of a thousand times larger than faculae [Dumusque et al., 2014] - on Figure 2, the variations could be dominated by several spots (and be considered as dips) or by large facular regions (peaks).

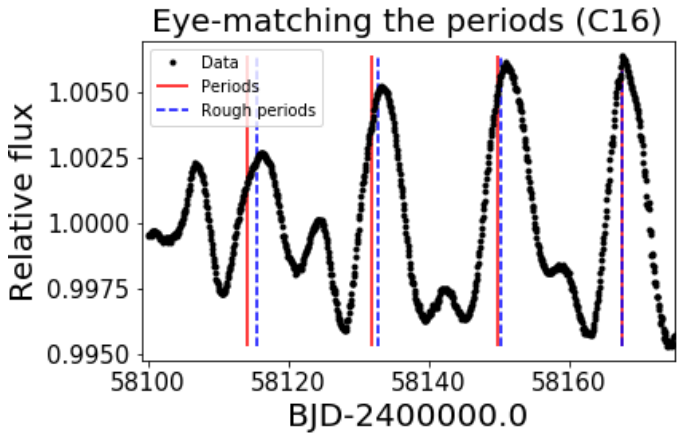

Figure 2: On this *K2* light curve, a pattern is clearly visible, caused by the rotation of the star. The period is computed and indicated with vertical lines. The two peaks forming the pattern, which are due to features on the two stellar hemispheres, change shape at each period as the features evolve.

1.1.2 Spectroscopy

In stellar spots on the photosphere, stronger magnetic fields inhibit convection. Therefore, the local blueshift due to rising material is suppressed. The reverse mechanism takes place with faculae. This will alter measurements of the receding and approaching velocities of the stellar hemispheres [Dumusque et al., 2014]; and therefore radial velocity measurements for exoplanet detection.

In current research, the Calcium II ion H and K emission lines in the ultraviolet range are seen as the main indicators of stellar activity. Indeed, these lines are produced in regions of the chromosphere (plages) that 'mirror' photospheric features. When selecting inactive stars for exoplanet surveys, the sample is probed for this doublet.

The process linking photospheric features and the appearance of chromospheric plages is not fully known, although it is believed that faculae are the main actors of this process, and generally surround spots. The many exceptions and subtleties of these rules call for further research.

1.2 Data: K2 photometry

Kepler is a space observatory trailing the Earth in an heliocentric orbit. It produces stellar photometric data (light curves) in the visible range. Observations are split into campaigns of about 80 days for each target field containing tens of thousands of stars.

The primary mission, which started in 2009 and greatly improved exoplanet research, was ended by the loss of two reaction wheels in 2013. *K2* was thus introduced as a way of continuing the use of the diminished spacecraft's capacities. In this mission, the telescope is pointed along the ecliptic, where solar pressure is minimal on its roll axis. The reaction wheels stabilise the other two axes, while the roll angle is corrected every few days with thruster firings [How-

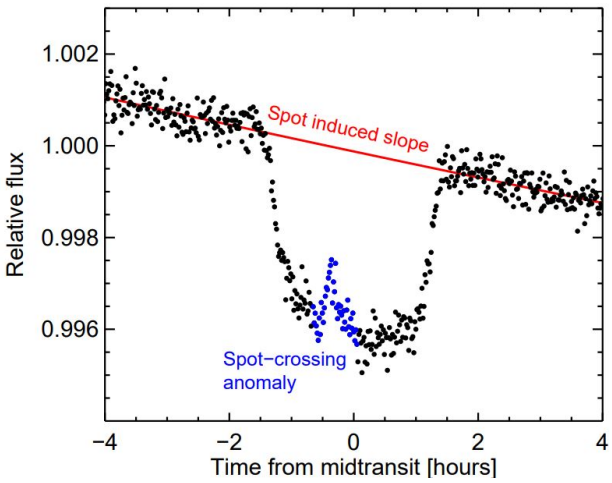

Figure 3: Example of two effects of activity features on a transit from Sanchis-Ojeda et al. [2013]. The slope is due to an active region moving in the field of view by stellar rotation - limb darkening making the change gradual (on Figure 2, limb darkening on each side of the rotating stellar disk leads to sinusoidal shapes). The peak is due to a sunspot disappearing behind the planetary disc towards the centre of the star - the base stellar flux is now higher, and the transit loses depth. Depending on the composition of the features and their position, several scenarios of transit modification can arise.

ell et al., 2014] (Figure 4). This instability in roll induces more difficulty in data cleaning and analysis.

In this project, long (30-minute) cadence data from *K2* Campaign 16 (December 2017 to February 2018) and 5 (April to July 2015) is analysed. The spacecraft was then in forward-facing mode (Figure 5), which allowed simultaneous observations from the Earth.

2 The frame of work

Dr. Alexandre Santerne at Marseille's Astrophysics Laboratory and his doctoral student Théo Lopez are participating in the global research effort to correct stellar activity in exoplanet detections. This is divided in two projects - a spectroscopic and photometric analysis of 83 stars (2.1), and a multi-wavelength and multi-technique in-depth study of four stars (2.2).

2.1 Large sample

This project studies the chromospheric Ca II H and K lines (1.1.2) together with stellar photometry to provide an insight into their relationship. For this purpose, the data was retrieved simultaneously: the spectra by the VLT's Ultraviolet and Visual Echelle Spectrograph, and the stellar light curves by *K2* (1.2). The proxy used is the $\log R'_{HK}$ indicator, which evaluates the true H and K flux using several spectral corrections.

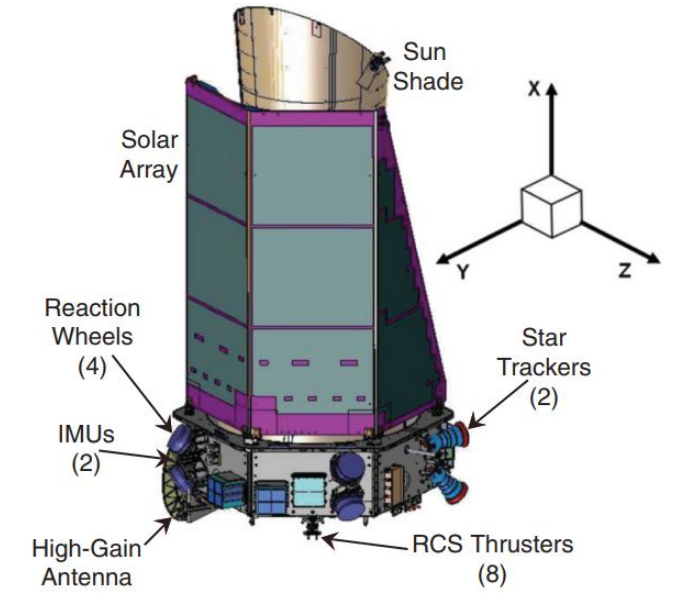

Figure 4: A representation of the axes of the *Kepler* space telescope from Howell et al. [2014]. The x (roll) axis is controlled periodically by thrusters, while the other two continuously by reaction wheels (which are a type of flywheel).

A dividing line between active and inactive stars has been determined empirically to fall at $\log R'_{HK} = -4.7$ ([Vaughan and Preston, 1980, Henry et al., 1996, Lockwood et al., 2007]). This value was associated to a division between spot-dominated, active young stars and faculae-dominated inactive older stars like the Sun, with low photometric variability [Lockwood et al., 2007]. Nevertheless, this inference could not then be verified as ground-based measurements of photometric variations are limited to about 1 mmag (the average atmospheric variation). Although photometric variability shows an increasing correlation with the chromospheric indicator above the Vaughan-Preston gap in Radick et al. [1998], one could only hypothesise its behaviour below this threshold (Figure 6).

The aim of this 'large sample' project is to explore this relationship below this limit, using *K2* space-based photometry, to ensure that the $\log R'_{HK}$ is a pertinent indicator of stellar noise in exoplanet detections.

There are a few scenarios that can arise:

- Stars below the Vaughan-Preston gap could get increasingly quiet with decreasing $\log R'_{HK}$ and not produce noise in transits or radial velocity measurements;
- Some stars that present no photometric variability could have cancelling features, and therefore only contaminate detections once the features have evolved. These stars could still be characterised as active using the $\log R'_{HK}$ indicator, but diverge from the linear relationship in Figure 6;
- The linear relationship between the $\log R'_{HK}$ and photometric variations

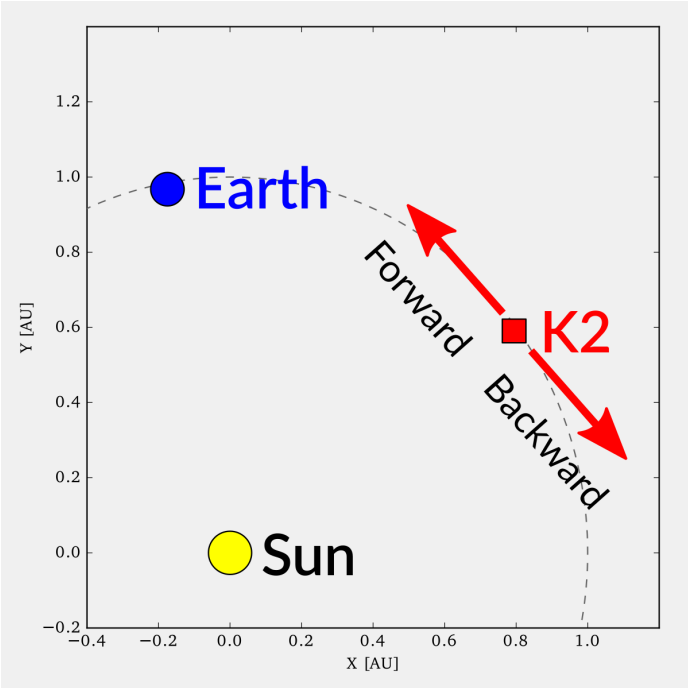

Figure 5: In forward-facing mode, simultaneous ground-based observations are possible (NASA/*Kepler* and *K2*/Geert Barentsen).

could not hold below the Vaughan-Preston gap due to degeneracy effects between spots and faculae, and a complex link between photosphere and chromosphere in this range.

It is essential to understand this relationship for the future of transit detections, including next-generation instruments such as CHEOPS, PLATO (ESA), and TESS (NASA). Their improved precision requires a trustworthy indicator of low stellar activity for sample selection.

This study focuses on bright Main Sequence stars less active than the Sun, in the F to M spectral types, that have been observed in two *K2* campaigns (5 and 16). The analysis on the log R'_{HK} indicator was done in a parallel internship described in Vivien [2018].

2.2 Small sample

This project (Théo Lopez's thesis) focuses on four stars to explore several physical relationships. In addition to the data of the large sample project, more spectroscopic indicators are considered, as well as polarimetry data. This will allow the study of:

- Radial velocity variations with respect to activity indices;
- Stellar magnetic mapping with respect to light curve shape, using light curve modelling;

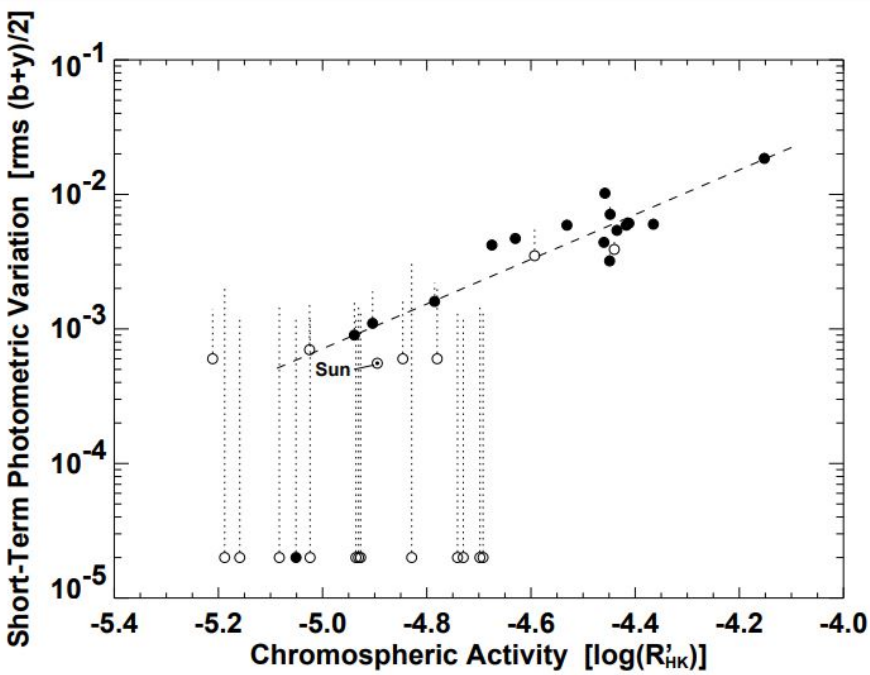

Figure 6: Relationship between photometric variation and the log R'_{HK} indicator from Radick et al. [1998]. The large error bars show the atmospheric limitation.

- Stellar activity variation with feature type (spot or faculae-dominated) and spectral class.

Simultaneous data is taken in spectroscopy with HARPS (visible), CARMENES (near infrared), SOPHIE (visible, long coverage), and in polarimetry with ESPaDoNS and NARVAL. The sample consists of four bright magnetic stars with a clear activity signal and a period fit for polarimetry.

3 The project

The aim of this internship was to retrieve the amplitude of variability of the sample's *K2* light curves, to understand their relationship with the log R'_{HK} indicator (2). Both stellar samples are studied, although a few stars are dismissed from the study:

- Stars that are indentified as binaries in Vivien [2018], as the studies focus on single stars;
- Stars for which no data was recorded during the *K2* campaign 16 or the UVES observations;
- Stars for which no good correction of instrumental noise was provided.

3.1 Computing rotation periods with the ACF method

3.1.1 Introduction

For a better familiarisation with the data and how to optimise it for fitting, the rotation periods of the sample were first studied. This later allowed a cross-check when fitting the curves which yields periods and amplitudes together. This is useful as retrieving a robust period should lead to a good fitted amplitude.

The standard approach to determining rotation periods from light curves is the periodogram, which uses Fourier decomposition or least squares fitting of sinusoidal models. However, these methods sometimes fail to encompass the complexity of stellar light curves. Indeed, although the latter are periodic, they have time-evolving features, undergo differential rotation, and despite corrections instrumental noise can remain important.

A good method to override these difficulties is the Auto-Correlation Function (ACF). It is a self-convolution (Figure 7) of the light-curve: the photometric data is overlapped with itself and shifted in time, and the area of overlap is retrieved one datapoint shift at a time. The ACF therefore measures the similarity of the overlapping curve to the initial curve at each shift.

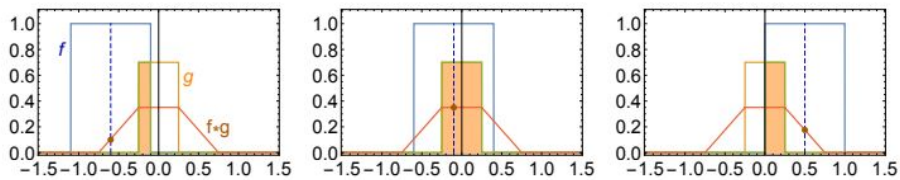

Figure 7: Convolution of two functions f and g. f is shifted along the x-axis and the evolving overlap is shown in orange. The value of the overlap at each shift, $f * g$, is the convolution. When performed with the same curve twice, $f * g$ is the ACF. (Dr. Sonja Franke-Arnold, Mathematical Methods for Physics, 2017).

The ACF of a periodic signal will hence show local maxima in similarity at multiples of the period (where the overlap is largest: two periodic patterns align). It is normalised and ranges between -1 and 1.

This method is presented in McQuillan et al. [2013]. The paper argues that the ACF is capable of producing more robust results than periodograms: the non-stellar noise and feature evolution is absorbed in the base level of the ACF, while local maxima will still show (Figure 8, 9).

3.1.2 Data cleaning

Each *K2* campaign lasting about 70 days, with readings done every 30 minutes, there are about 3000 datapoints per star. Pre-search Data Conditioned Simple Aperture Photometry (PDCSAP, Smith et al. [2017]) is the first coarse correction of the raw light curves provided by the *Kepler* team. It aims to correct for optical artifacts such as thermally-driven focus changes, pointing drifts, cosmic rays, and relativistic aberration (Figure 10). They also correct interruptions in the data due to crashes and Earth downlinks, causing large jumps. The main method used in Smith et al. [2017] is the removal of correlation between

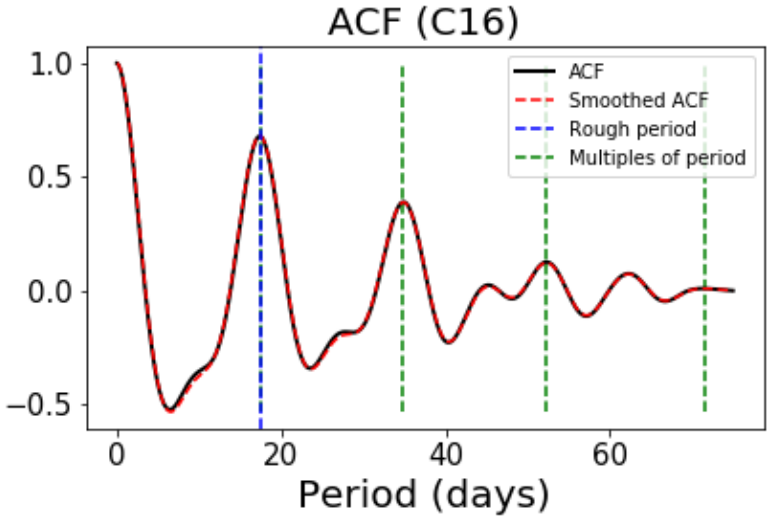

Figure 8: On this ACF of the light curve presented in Figure 2, the marked peaks correspond to period multiples. The light curve pattern in 2 has two peaks per period: one large peak and one smaller peak, due to different features in the two stellar hemispheres. When the large peak overlaps the other in the ACF, a similarity is detected, but it is not as significant as when the full periodic pattern overlaps itself. If all the local maxima of the ACF were chosen, the deducted period would be of about half the true value, as one pattern would be attributed to two rotations of the star. Note: at the start, the light curve is in perfect alignement with itself, giving a self-similarity of 1, which decreases rapidly. The details of peak selection are detailed in 3.1.3.

light curves and the elimination of long-term trends, with special care taken to preserve stellar variability.

A second correction needs performed that focuses on the pointing drift. There are several 'High-Level Science Products' (HLSPs), approved by NASA, that provide corrected light curves. There was only two HLSPs that preserved stellar variability (most remove it for transit detection) and that had already corrected Campaign 16 data during the internship: 'Planet candidates for Optimal Aperture reduction', POLAR [Barros et al., 2016]; and '*K2* Self Flat-Fielding', K2SFF [Vanderburg and Johnson, 2014]. Their difference is that K2SFF uses the pipeline aperture mask (pixels chosen for data collection of a given star) while POLAR generates its own using a CoRoT routine.

The pointing drift systematics come from the telescope rolling about its line of sight (1.2). The star moves in an arc about this roll axis in the CCD, and goes back to its original position with thruster firings every few days. The main culprit is pixel sensitivity variation, and therefore simply making a large aperture mask to avoid losses is not adapted. As a matter of fact, creating an aperture mask following the star degrades the data [Barros et al., 2016]. It is therefore more fit to leave this centroid variation and correct for sensitivity fluctuation jitter. There are also a few remaining longer-term trends that slipped through PDCSAP.

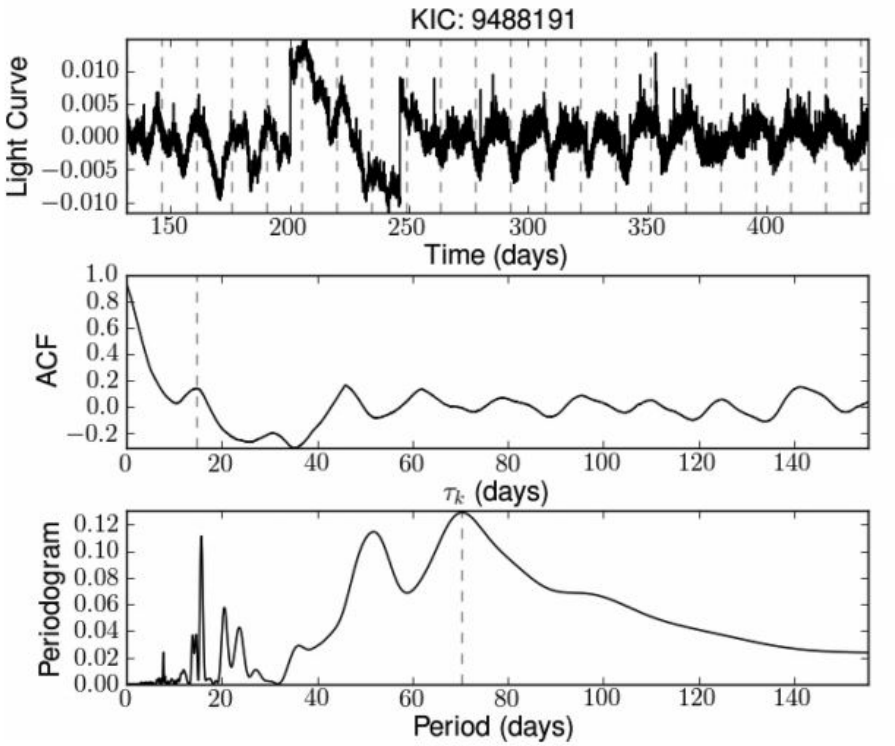

Figure 9: In this comparison from McQuillan et al. [2013], the least-squares fitting periodogram does not compute the correct stellar rotation period due to instrumental effects (notably in days 200 to 250). The systematic trend is visible in the ACF but local maxima are still present at each period multiple.

The method applied to correct for the sensitivity variations is the Self Flat-Fielding method, introduced in Vanderburg and Johnson [2014]. It correlates position variation and flux variation.

It was observed that POLAR was more detrended than K2SFF - and therefore it was chosen for most light curves, except for three where the correction failed. Campaign 5 PDCSAP was an older and thus poorer correction, leaving most of the large jumps due to downlinks and crashes. Consequently, validation of periods using a second light curve was mostly not done.

In this project's algorithm, a few further corrections were performed: outliers, NaNs and remaining relativistic long-term trends were removed (for the latter, by second degree polynomial fitting); and the data was normalised to 1. The SFF correction can cause slopes at the start and end of the dataset, and so did the Earth being near the field of view at the beginning of Campaign 16. Those were removed after visual inspection.

3.1.3 Algorithm

Computing the ACF The python package `interpacf` by Brett Morris reproduces some of the ACF algorithm in McQuillan et al. [2013]. Using it, the

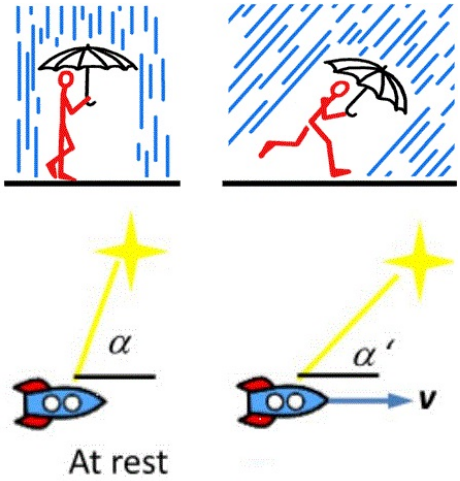

Figure 10: Relativistic aberration effect in optics, where $\tan\frac{1}{2}\alpha' = \frac{c-v}{c+v}\tan\frac{1}{2}\alpha$ (Dr. Thomas Brougham, Special Relativity for Astronomy, 2016).

lightcurve was first interpolated to have regular cadences (which are needed to allow the self convolution), and the ACF was computed. To avoid the detection of very short periods (e.g. centroid variation jitter), the ACF was smoothed by convolution with a Gaussian.

Peak selection and finding the period First, a rough estimate of the period was made by locating the first period peak (the 'rough period') in the ACF. The ACF maxima with the highest relative height in the first two was selected (to avoid half periods, Figure 8). The half width at half maximum of this peak was defined as the rough period uncertainty.

Then, the other period peaks were found with the following criteria applied on all the maxima of the ACF [McQuillan et al., 2013]:

- The peak must be within 20% of an integer multiple of the rough period;
- The peak is consecutive to another.

No more than 10 peaks were selected. The median of the time interval between the sample of maxima was retrieved as a more robust period estimate. The error was calculated using:

$$\alpha_{period} = \frac{1.483 \cdot |\tilde{\delta_i - \tilde{\delta}}|}{\sqrt{N-1}} \tag{1}$$

where N is the number of selected peaks, δ_i the time intervals between them, and tildes represent medians [McQuillan et al., 2013].

The period, rough period and their error were then visualised by tracing vertical lines on the light curves, taking the maximum as a reference point. Lines were also set at the dates of UVES observations to visualise at what stage of activity the spectrum was taken.

3.2 Gaussian Processes

3.2.1 Introduction

There are drawbacks to the ACF method [Angus et al., 2017] – it needs data to be evenly spaced and does not generate trustworthy errors. It also requires a few harsh choices, like the parameters of the smoothing Gaussian (Figure 11). Moreover, picking the highest of the first two ACF peaks is not always reliable.

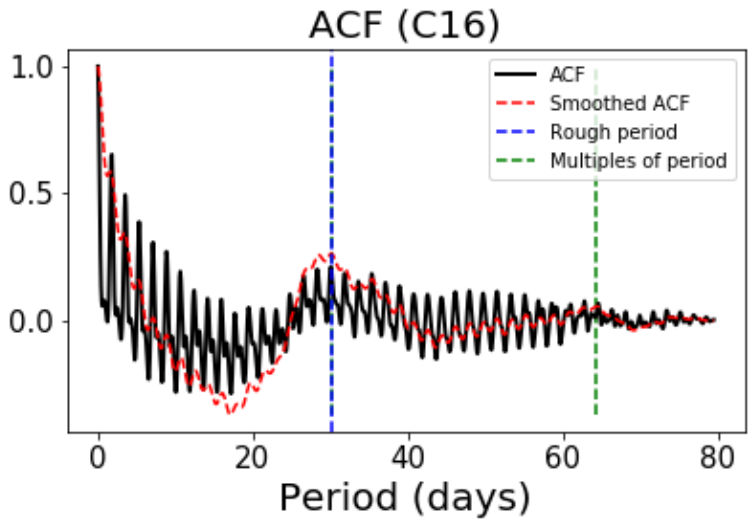

Figure 11: An ACF case of over smoothing: the short, clear period has been ignored.

Gaussian Processes (GPs) are a modelling technique - here the light curves are fitted unlike the ACF process. They model correlation in the data rather than the data directly. A kernel is chosen to model the covariance structure, i.e. the 'variation in correlation as a function of separation between data points' [Angus et al., 2017]. Here the kernel used was the quasi-periodic (QP) kernel, which relates the time and flux axes on the light curve:

$$kernel = c \cdot \exp[\frac{-\Delta t^2}{\lambda_1^2} - \frac{2\sin^2 \frac{\pi|\Delta t|}{P}}{\lambda_2^2}] \tag{2}$$

with c the amplitude of variability, P the stellar rotation period, λ_1 the feature decay time, and λ_2 a weighting of the sinusoidal component. The GP can be described by:

$$p(\theta|D, H, I) \propto \mathcal{L}(D|\theta, H, I) \cdot P(\theta|H, I) \tag{3}$$

with:

- p the posterior probability density distribution of the parameters θ (the results);
- $\mathcal{L}$ the likelihood of the fit;

- D the data (the light curve);
- H the hypotheses made (e.g. 'the amplitude of variability is anywhere between 0 and 1'),
- and I the information available (e.g. a distribution of stellar periods from a previous paper);
- and P the priors (all the limits and distributions given to parameters), which is a combination of H and I.

The GP therefore provides a nearly complete physical model of the star and generates robust errors. The priors and choice of kernel are the subjective choices of this method. For more literature on GPs, Roberts et al. [2013] provides a detailed review of the method.

3.2.2 Data cleaning

The main downside of Gaussian Process is that they are expensive computationally. They consist of a Monte Carlo Markov Chain (MCMC), with about 100 000 steps, to test the likelihood and converge to QP parameters. About ten chains have to be run to eliminate local maxima of the likelihood.

The data was rebinned to a median time and flux every six hours, and the new flux error (photon noise) of this downsampled light curve was calculated (median flux error divided by square root of number of points) to reduce computing time.

3.2.3 Algorithm

The software used for this GP analysis was `PASTIS` (Planet Analysis and Small Transit Investigation Software, Díaz et al. [2014]). It contains a module specific to *K2* photometry, written by Alexandre Santerne. The priors were optimised with three 3-day test runs, and finally set as follows:

- Foot: uniform distribution between 0.9 and 1.1, with a random start of the chain. This is the 'base' of the light curves, which were all normalised to 1;
- Contamination: the amount of noise brought in by other stars in the aperture mask. There was no data for this, so it was set and fixed to 0;
- Jitter: uniform distribution between 0 and 1, the noise caused by the centroid movement;
- Amplitude: uniform distribution between 0 and 1;
- Period: Gaussian centred at 20 days, with a width of 10 days, and truncated at 3 and 40 days. It was taken from McQuillan et al. [2014], where the ACF was performed on 34 000 *K2* light curves;
- λ_1 (feature decay time): uniform distribution between 3 and 500 days [Giles et al., 2017];
- λ_2 (weight of sine component): uniform distribution limited to 10, to avoid the disappearance of the sine constituent.

These choices were checked at each run by plotting their influence of parameters and results; and the final set did not seem to constrain or limit them. For comparison, Angus et al. [2017] put a tighter constraint on their Gaussian process in similar work. They cleaned up the data with bandpass filters eliminating unwanted periods, left less freedom on λ_2 to avoid both extremes, and used the ACF to create loose bounds on the period of each star.

3.3 Results and analysis

3.3.1 Periods and amplitudes

The retrieved periods are plotted together on Figure 12. It can be suggested that the Gaussian process results are superior to the ACF's for this sample:

- Fast rotators are detected by the GP, but not by the ACF due to its smoothing. They are placed around the 4-day period line, and their high amplitude can differentiate them from truly inactive stars, where only noise remains. This was an unfounded concern for the Gaussian Process: it was thought probable that the algorithm would catch small jitter 'periods' over a real rotational period since there no smoothing was performed.
- The line $y = 2x$ seems to indicate that the fit finds more double peak patterns. The stars of the sample have a low-amplitude of activity, which can make jitter and activity difficult to differentiate, so this ambiguity is expected. One may trust the Gaussian process more on this choice since the data is fitted with a model, closer to physical reality.
- Finally, a number of results are similar or identical ($y = x$), which is a good sign of both methods functioning on clear signals.

The amplitudes retrieved with the Gaussian process fit were very similar to a root mean square of the corrected light curves (Figure 13.

3.3.2 Amplitude of variability and chromospheric activity indicator relationship

On Figure 14 are presented the final results of this study. The $\log R'_{HK}$ is taken from Vivien [2018]. An increasing trend with a large spread appears, which confirms a relationship between the two quantities as hypothesised in Radick et al. [1998], although quite loosely.

The higher amplitude stars are probably spot dominated, as the latter introduce more variability [Dumusque et al., 2014]. They spread throughout the whole range of $\log R'_{HK}$, meaning the relationship does not seem to hold very well for spot-dominated stars. Spots not heating the chromosphere as much as faculae could justify the presence of these variable, low $\log R'_{HK}$ stars in the upper left corner of the diagram.

On the other hand, a high $\log R'_{HK}$ maps to higher amplitudes. This allows removal of feature-rich stars (with both spots and faculae, the latter causing most of the Ca II H and K emission) from exoplanet research samples.

In the lower left corner is a large group of truly inactive stars, which justifies the use of the chromospheric indicator. They seem to have higher periods, which is mostly due to no activity appearing in the light curves and no true rotational period being detected (the Gaussian prior centred on 20 days, then dominates).

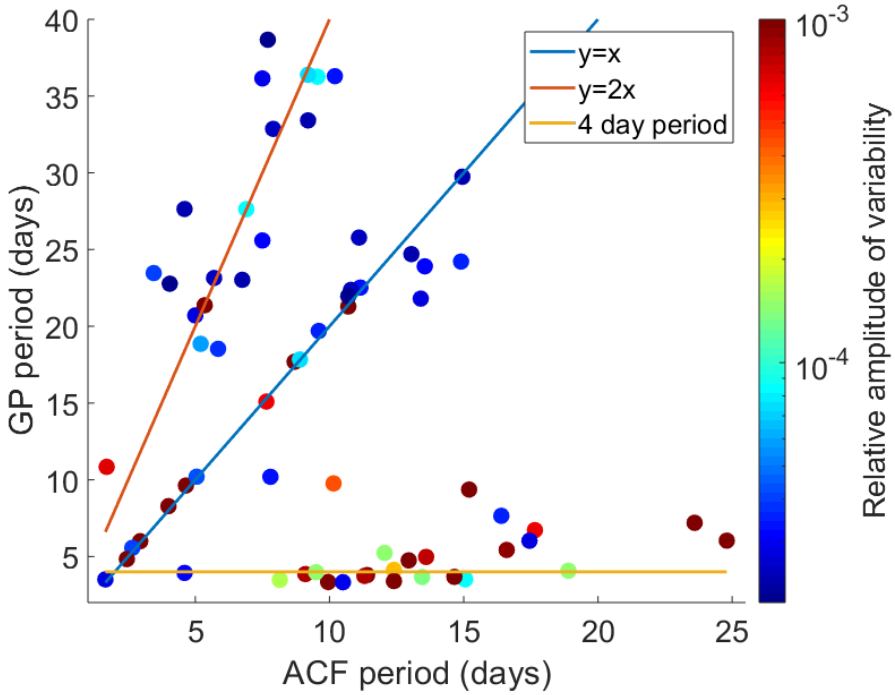

Figure 12: A comparison of the periods retrieved by the ACF and Gaussian processes. The Gaussian process may be able to detect double peak patterns better than the ACF, for which half the period is detected (line $y = 2x$) and fast rotators (which are mainly not jitter as they have high amplitude).

3.4 Discussion and openings

3.4.1 Improvements

There is a number of additions that could have been performed to increase the relevance of this project:

- Smoothing the ACF differently depending on the light curve, catching both short periods and larger periods in noisy, low-amplitude light curves;
- An improvement of the detection of peaks in the ACF to leave out irregularities due to noise;
- If the second maxima was chosen as the period in the ACF, checking that the even peaks are larger than their neighbours would have been safer. This would have secured the detection of a repeating double pattern and brought the results of the Gaussian Processes and ACF closer;
- Allowing shorter periods than 3 days in the Gaussian Process prior to reduce subjectivity.

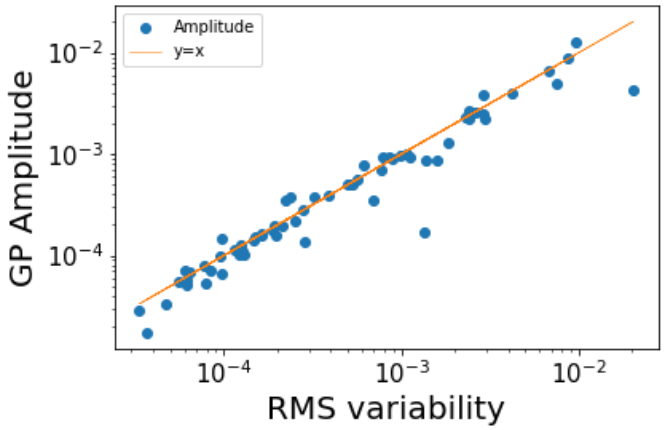

Figure 13: A comparison of the Gaussian Process fitted amplitudes and a simple root mean square of the light curves. Although similar, the slight differences justify the use of a fit.

3.4.2 Further research

This project has opened a number of new leads to look into, but on the other hand, some issues have yet to be solved.

There are several light curves for which standard period retrieval does not apply. On Figure 15 are such cases: at the top, a star which could possibly be almost pole on (a feature near the visible pole could almost always be in view, causing the large trends; and one near the hidden pole cause the quicker variations); and on the bottom curve, a star is both pulsating and rotating.

Further, issues have arised with the Gaussian Processes: some fits have not been performed, leaving most parameters at zero, and errors were very large in general. Although the results were satisfactory, they were unconstrained. Further research is needed to improve their quality.

To test the quality of the results, rotational velocities can be derived from rotation periods and plotted against FWHMs from Vivien [2018] or temperatures from the *GAIA* mission. Additional investigation can also be done with the data retrieved: looking at the variation of the chromospheric indicator in the CA II infrared IRT lines [Vivien, 2018] with amplitude of variability, or splitting Figure 14 by spectral type.

Finally, Campaign 5 *Kepler* photometry could be corrected to provide a better period prior for each star.

4 Conclusion

A large part of the time dedicated to this project was spent on rotational periods. It has been found that both methods applied to retrieve them have limitations as they require subjective choices. Although heavier on computational time, the Gaussian Process presented more relevant results overall.

The main conclusion of this project was that the $\log R'_{HK}$ indicator does

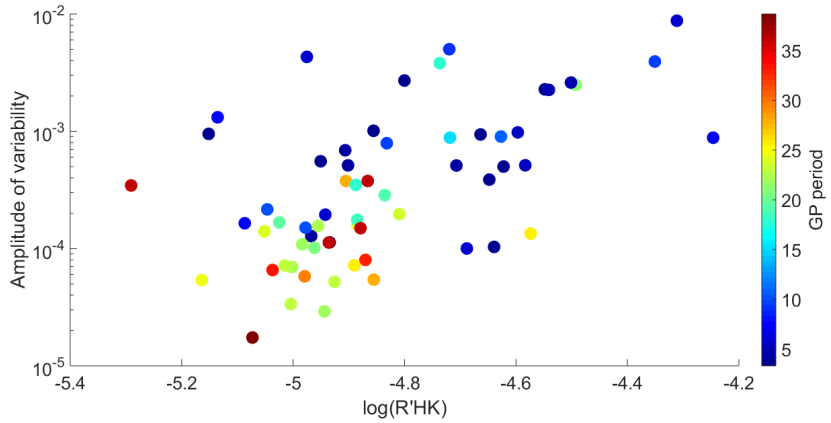

Figure 14: The increasing linear trend hypothesized in Radick et al. [1998] (Figure 6) seems to hold quite loosely for low $\log R'_{HK}$. Spot dominated stars, for which there is little chromospheric heating and high amplitude of variability, create a spread of the relationship towards the upper left corner of the graph.

not have a simple linear relationship with photometric variability. Low $\log R'_{HK}$ stars are not all truly quiet and can have features that produce noise in transits or radial velocity data. This is a problem for next-generation instruments that select their sample of 'inactive' stars for very precise measurements using this indicator. Further research is undertaken at the LAM's GSP to better understand stellar activity (2.2).

References

Ruth Angus, Timothy Morton, Suzanne Aigrain, Daniel Foreman-Mackey, and Vinesh Rajpaul. Inferring probabilistic stellar rotation periods using Gaussian processes. *Monthly Notices of the Royal Astronomical Society*, 474(2):2094–2108, jun 2017. URL http://arxiv.org/abs/1706.05459.

S. C. C. Barros, O. Demangeon, and M. Deleuil. New planetary and EB candidates from Campaigns 1-6 of the K2 mission. *Astronomy & Astrophysics*, 100:1–9, 2016. URL http://arxiv.org/abs/1607.02339.

R. F. Díaz, J. M. Almenara, A. Santerne, C. Moutou, A. Lethuillier, and M. Deleuil. Pastis: Bayesian extrasolar planet validation-I. General framework, models, and performance. *Monthly Notices of the Royal Astronomical Society*, 441(2):983–1004, 2014. URL https://arxiv.org/abs/1403.6725.

X. Dumusque, I . Boisse, and N. C. Santos. SOAP 2.0: A tool to estimate the photometric and radial velocity variations induced by stellar spots and plages. *The Astrophysical Journal*, 796(2):132, sep 2014. URL http://arxiv.org/abs/1409.3594.

Helen A. C. Giles, Andrew Collier Cameron, and Raphaëlle D. Haywood. A Kepler study of starspot lifetimes with respect to light-curve amplitude and

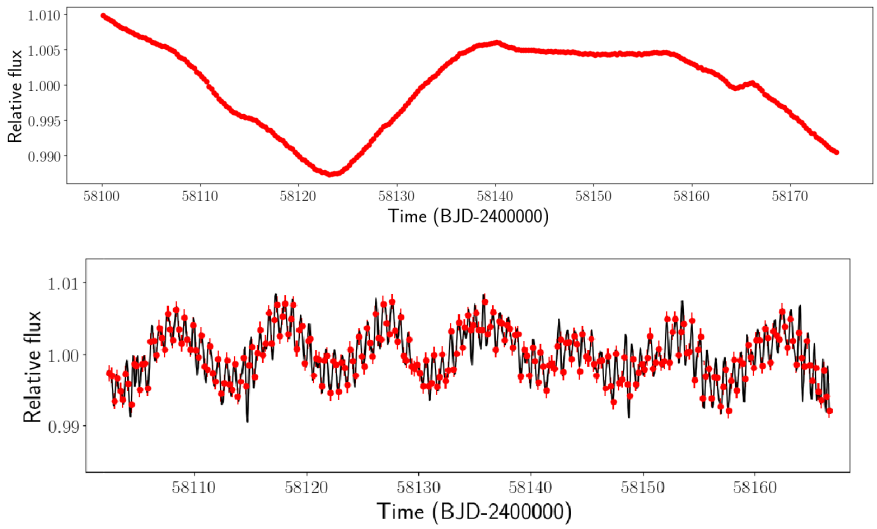

Figure 15: Two peculiar light curves from the sample.

spectral type. *Monthly Notices of the Royal Astronomical Society*, 472(2): 1618–1627, dec 2017. URL https://arxiv.org/abs/1707.08583.

Todd J. Henry, David R. Soderblom, Robert A. Donahue, and Sallie L. Baliunas. A Survey of CA II H and K Chromospheric Emission in Southern Solar-Type Stars. *The Astronomical Journal*, 111:439, jan 1996. URL http://adsabs.harvard.edu/cgi-bin/bib_query?1996AJ....111..439H.

Steve B. Howell, Charlie Sobeck, Michael Haas, Martin Still, Thomas Barclay, Fergal Mullally, John Troeltzsch, Suzanne Aigrain, Stephen T. Bryson, Doug Caldwell, William J. Chaplin, William D. Cochran, Daniel Huber, Geoffrey W. Marcy, Andrea Miglio, Joan R. Najita, Marcie Smith, J. D. Twicken, and Jonathan J. Fortney. The K2 Mission: Characterization and Early results. *The Astronomical Society of the Pacific*, 128(938):398–408, 2014. URL http://arxiv.org/abs/1402.5163.

G W Lockwood, B A Skiff, Gregory W Henry, Stephen Henry, R R Radick, S L Baliunas, R A Donahue, and W Soon. Patterns of Photometric and Chromospheric Variation Among Sun-Like Stars: a 20 Year Perspective. *The Astrophysical Journal Supplement Series*, 171(1966):260–303, 2007. URL https://arxiv.org/abs/astro-ph/0703408.

A. McQuillan, S. Aigrain, and T. Mazeh. Measuring the rotation period distribution of field M dwarfs with kepler. *Monthly Notices of the Royal Astronomical Society*, 432(2):1203–1216, 2013. URL https://arxiv.org/abs/1303.6787.

A. McQuillan, T. Mazeh, and S. Aigrain. Rotation periods of 34,030 Kepler main-sequence stars: The full autocorrelation sample. *Astrophysical Journal, Supplement Series*, 211(2), 2014. URL https://arxiv.org/abs/1402.5694.

Richard R. Radick, G. W. Lockwood, B. A. Skiff, and S. L. Baliunas. Patterns of Variation among Sun-like Stars. *The Astrophysical Journal Supplement Series*, 118(1):239–258, 1998. URL http://stacks.iop.org/0067-0049/118/i=1/a=239.

S. Roberts, M. Osborne, M. Ebden, S. Reece, N. Gibson, and S. Aigrain. Gaussian processes for time-series modelling. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 371(1984):1–27, 2013. URL http://rsta.royalsocietypublishing.org/content/371/1984/20110550.

Roberto Sanchis-Ojeda, Joshua N. Winn, Geoffrey W. Marcy, Andrew W. Howard, Howard Isaacson, John Asher Johnson, Guillermo Torres, Simon Albrecht, Tiago L. Campante, William J. Chaplin, Guy R. Davies, Mikkel N. Lund, Joshua A. Carter, Rebekah I. Dawson, Lars A. Buchhave, Mark E. Everett, Debra A. Fischer, John C. Geary, Ronald L. Gilliland, Elliott P. Horch, Steve B. Howell, and David W. Latham. KEPLER-63b: A giant planet in a polar orbit around a young sun-like star. *Astrophysical Journal*, 775(1), 2013. URL https://arxiv.org/abs/1307.8128.

Jeffrey C Smith, Martin C Stumpe, Jon M Jenkins, Jeffrey E Van, Forrest R Girouard, Jeffery J Kolodziejczak, Sean D Mccauliff, Robert L Morris, and Joseph D Twicken. Presearch Data Conditioning. *Kepler Data Processing Handbook*, KSCI-19081:159–235, 2017. URL https://archive.stsci.edu/kepler/manuals/KSCI-19081-002-KDPH.pdf.

Andrew Vanderburg and John Asher Johnson. A Technique for Extracting Highly Precise Photometry for the Two-Wheeled Kepler Mission. *PASP*, 126 (944):948–958, oct 2014. URL https://arxiv.org/abs/1408.3853.

Arthur H Vaughan and George W Preston. A survey of chromospheric CA II H and K emission in field stars of the solar neighborhood. *Publications of the Astronomical Society of the Pacific*, 92(August):385, aug 1980. URL http://iopscience.iop.org/article/10.1086/130683.

Hugo Vivien. Exploring the validity of the log(RHK) as an activity indicator in main-sequence stars. 2018.